Comprehensive and streamlined quality control for single-cell RNA data with singleCellTK

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INTRODUCTION

- The prevalence of single-cell RNA-sequencing (scRNA-seq) has caused a rapid increase in available data as well as a proliferation of computational tools used for quality control and analysis, which has resulted in a need for the creation of a streamlined protocol for conducting scRNA-seq analysis.
- Performing quality control is necessary to remove poor quality cells as well as artifacts in the scRNA-seq data, such as doublets or ambient RNA, which can confound downstream analyses or produce erroneous finding.

MAJOR FEATURES

- The singleCellTK QC (SCTK-QC) pipeline is able to import data from various preprocessing tools (e.g. 10X CellRanger) and export the result in various format to accommodate downstream analysis in different programming language.
- The SCTK-QC integrates numerous tools for comprehensive quality control such as generating standard QC metrics, detecting empty droplet, doublet and ambient RNA contamination
- The singleCellTK incorporates Shiny graphical user interface for interactive analysis of single-cell data
- We have developed a Unix script to run the pipeline as command-line tool. The pipeline was built into docker image and being deployed on Terra platform.

METHODS

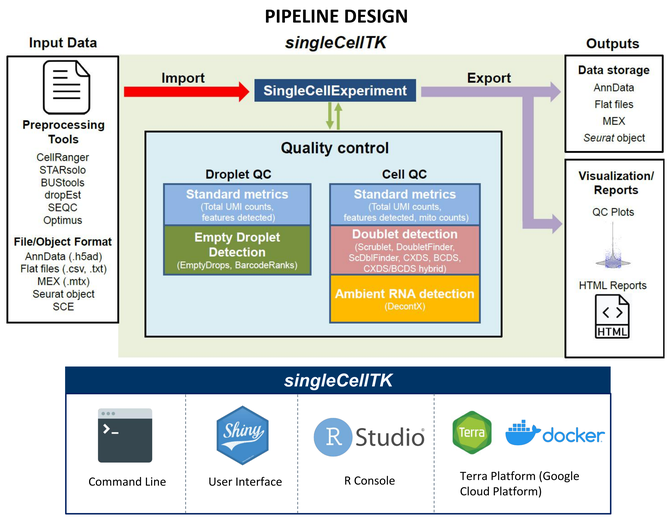

Figure 1. The SingleCellTK QC pipeline is developed in the R environment and can take in as input datasets generated from preprocessing tools such as *CellRanger* from 10X Genomics. Quality control, which includes calculation of general quality control metrics and the detection of doublet and ambient RNA, as well as data visualization, and report building can be subsequently conducted on the imported dataset based on user specified parameters. Within SingleCellTK, the data is stored as a SingleCellExperiment object, which may be exported as a R Seurat object, Python AnnData object, or a .txt flat file. The above workflow may be conducted straight from the command line, from user interface, from the R console, or from cloud computation platform (Terra).

RESULT

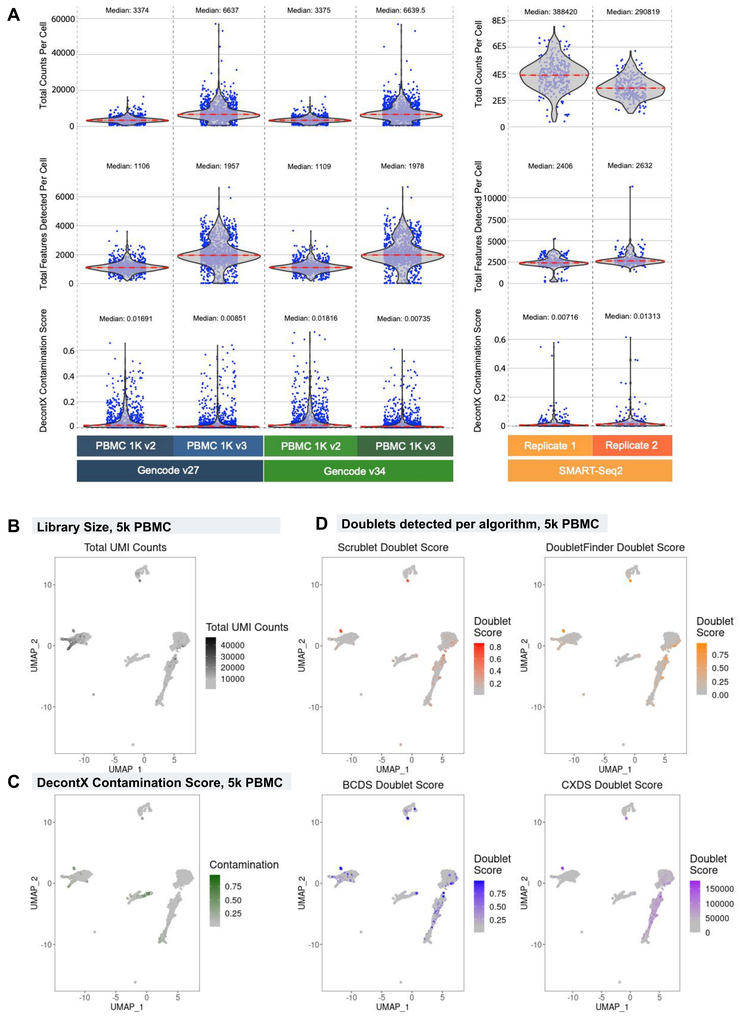

Figure 2. (A) SingleCellTK utilizes the *scran* package to compute general quality control metrics. The number of UMI counts (top) , the number of features detected (middle) and ambient RNA contamination score (bottom) were determined for PBMC 1K V2, PBMC 1K V3 from 10x Genomics and two replicate human PBMC dataset generated with SMART-Seq2 protocol, available from Human Cell Altas Single Cell Portal. **(B)** UMAP was applied on the 10X 5K PBMC (V3) dataset for dimensionality reduction, and shaded by the overall library size. **(C)** DecontX computes the level of contamination caused by ambient RNA per cell. The contamination score per cell is visualized. **(D)** SingleCellTK employs several doublet detection algorithms to compute doublet scores, which corresponds to the likelihood that a particular cell droplet is a doublet. The doublet score for the 10X 5K PBMC (V3) dataset is shown. (Top left: Scrublet, Top right: DoubletFinder, Bottom left: BCDS, Bottom Right: CXDS).

GENERATING REPORTS

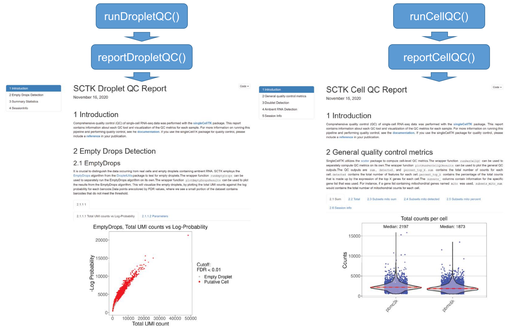

Figure 3. SingleCellTK generates the reports with the analysis results in the .html format. The fragment of the DropletQC and CellQC report is demonstrated. reportDropletQC() generates a report including a scatterplot annotating all empty droplets flagged by the EmptyDrops algorithm as well as a curve visualizing the knee and inflection points identified by BarcodeRanks. For each set of doublet detection algorithm executed, reportCellQC() generates a report which visualizes the doublet score and call as well as contamination percentage of ambient RNAs through violin plots, density plots, and dimensionality reduction plots.

CONCLUSIONS

- The SCTK-QC pipeline streamlines and standardizes the generation and visualization of metrics important for assessing data quality.
- SCTK-QC is able to generate HTML reports with publication-ready figures and contains a GUI for interactive QC of single-cell data. These features will enable users without in-depth programming backgrounds to run these tools and perform QC on their data.
- The modular architecture of SCTK-QC allows for quick and easy integration of any additional tool that may be available in the future.
- The SCTK-QC pipeline can export results to both R and Python-compatible data structures enabling easy integration with other popular analysis frameworks such as Seurat and Scanpy.

FUTURE DIRECTIONS

- Currently, the singleCellTK accommodates only scRNA-seq data. However, as single-cell multi-omic approaches become more popular, the ability to accommodate multiple types of single cell data will become imperative. We plan to add functionalities for other methods such as scATAC-seq, and CITE-seq.
- The singleCellTK is going to incorporate more tools for single-cell data analysis. Please reach out to us if you are developing tools and want to test it on our framework.
- Additionally, we plan to incorporate the singleCellTK user interface on the cloud environment for a better user experience.

LINKS

- R package: https://github.com/compbiomed/singleCellTK
- Documentation: http://sctk.camplab.net/v2.2.0/articles/articles/cmd_qc.html
- Campbell lab website: http://www.camplab.net/

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